SLIDE 1: TITLE PAGE: MPLADS-SATARK (8$0M)

AI Risk Scoring, Multi-Layer Anomaly Detection & Explainable Vigilance Engine

Problem Statement ID: 26102

Problem Statement Title: Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Sch

Ministry / Organization: Ministry of Statistics & Programme Implementation (MoSPI / DIID)

Theme: Smart Automation / Clean & Transparent Governance

PS Category: Software

Team Name & ID: [Insert Your Registered Team Name & Team ID]

Guidance: Clean, authoritative title slide with Government of India MoSPI branding and your team credentials.

SLIDE 2: PROPOSED SOLUTION: MPLADS-SATARK: Proactive Multi-Layer

[DETAILED EXPLANATION OF PROPOSED SOLUTION]

* Automated vigilance platform transforming passive historical records into real-time fraud prevention before public funds are disbursed.
* Evaluates project proposals and vouchers across 6 independent AI and forensic detection signals.
* Assigns a 0 to 100 Priority Risk Score with an itemized, additive Explain-My-Score breakdown citing exact General Financial Rules (GFR 2017).

[HOW IT ADDRESSES THE CORE PROBLEMS]

* Double-Billing: Matches works against ISRO Bhuvan / MGNREGA to prevent paying twice for the same road/hall.
* Overpricing: Calibrates costs against CPWD Delhi Schedule of Rates (DSR) to flag inflated estimates.
* Contractor Cartels: Network graph analysis exposes single-agency monopolies and bidding syndicates.
* Zombie Projects: Survival hazard modeling flags stalled works 6 months before physical deadlines lapse.

[INNOVATION AND UNIQUENESS]

* 100% Real Government Data: Validated on 176,925 works and 109,521 vouchers directly from live MoSPI endpoints.
* Honest Data Provenance: Immutable badges (REAL, DERIVED, ESTIMATED, STATUTORY) on every number. Zero black-box AI.
* 1-Click Printable Audit Dossier: Official Inspection Memorandum ready for field verification by District Engineers.

SLIDE 3: TECHNICAL APPROACH: Bifurcated Architecture & 6-Signal Detec

[TECHNOLOGIES & FRAMEWORKS]

* Backend: Node.js (v24), High-Performance In-Memory Columnar Store (sub-second query) ' (over 1.76L works), REST API.
* Frontend: Modern Vanilla CSS design system, Leaflet GIS geospatial mapping, Vis.js interactive network graphs, Chart.js.
* Statistical ML Models: Tree-based Isolation Forest, Louvain Community Detection, Benfords Law (Chi-Square), Cox Hazard Regression.

[METHODOLOGY & IMPLEMENTATION FLOW]

* Cold Path: Ingests MoSPI, CPWD DSR, and Bhuvan data offline into indexed cubes for instant 4-dashboard rollups.
* Hot Path (Live Demo): Scores newly injected project tenders in 15 milliseconds (exceeding 50ms SLA).
* 6-Signal Multi-Layer Defense: Combines Cost Variance (Sig 1), Network Cartels (Sig) ' (2), Statutory Rules (Sig 3), Cross-Scheme Ded and Survival Stalls (Sig 6).

[WORKING PROTOTYPE STATUS]

* Fully functional prototype live on localhost:3000 with 4 role-based dashboards (Ministry, State, Collector, MP).

SLIDE 4: FEASIBILITY AND VIABILITY: Production Feasibility, Risk Analysis

[FEASIBILITY ANALYSIS AT GOVERNMENT SCALE]

* Demonstrated Nationwide Scale: Successfully processed 100% of India (36 States) ’ (UTs, 541 Constituencies, 176,925 works) in und
* No Costly Infrastructure: In-process analytical engine operates without heavy
  distributed clusters or third-party paid APIs.

[POTENTIAL CHALLENGES & RISKS]

* Challenge 1 (Scale Risk): Large datasets and 27,000-node network graphs freezing
  web browsers during live demos.
* Challenge 2 (Changing Rules): Statutory policy guidelines (GFR / MPLADS quotas)
  changing over time, breaking hardcoded logic.
* Challenge 3 (Fuzzy Data Quality): Unstandardized Gram Panchayat spellings across
  MoSPI and MGNREGA.

[STRATEGIES FOR OVERCOMING CHALLENGES]

* Mitigation 1 (Focus Context Ego-Graphs): Renders 1-2 hop neighborhoods (10-15) ’ (nodes) on demand instead of thousands.
* Mitigation 2 (Policy-as-Data): Externalized bitemporal rules with effective date
  ranges past audits remain legally valid.
* Mitigation 3 (Block-Level Blocking): Hierarchical matching: State + District block,
  followed by n-gram token similarity.

SLIDE 5: IMPACT AND BENEFITS: Transformative Governance & Economic

[POTENTIAL IMPACT ON TARGET STAKEHOLDERS]

* Union Ministry / DIID: Real-time macro visibility into 7,908 Cr annual funds and cross-state utilization velocity.
* District Magistrate / Collector: Prioritized inspection queue reduces manual audit burden by 85%.
* Honble Member of Parliament: Proactive self-check dashboard prevents unintentional quota non-compliance.
* Citizens Taxpayers: Radical transparency ensuring sanctioned roads, hospitals, and drinking water actually exist.

[QUANTIFIABLE BENEFITS (ECONOMIC & SOCIAL)]

* Economic: Prevents an estimated 500+ Crore in annual fund leakages (overpricing,) ' (double-invoicing, March rush waste).
* Social Justice: Guarantees statutory 15.0% SC and 7.5% ST grassroots development quotas through real-time tracking.
* Constitutional Realism: Strict Human-in-the-Loop design AI assists officers with evidence but never freezes funds unilaterally.

SLIDE 6: RESEARCH AND REFERENCES: Ground Truth Citations & Resear

[PRIMARY GOVERNMENT DATASETS & PORTALS]

* Ministry of Statistics Programme Implementation (MoSPI): eSAKSHI Live Portal (https://mplads.mospi.gov.in/).
* Central Public Works Department (CPWD): Delhi Schedule of Rates (DSR 2023-24) State Cost Multipliers.
* ISRO / Ministry of Rural Development: Bhuvan Geo-Platform MGNREGA Rural Asset Registry.

[STATUTORY & LEGAL FRAMEWORKS]

* Ministry of Finance: General Financial Rules (GFR 2017) Rule 62 (Pacing), Rule 99 (Vouching), Rule 130 139.
* MoSPI: Revised MPLADS Scheme Guidelines 2023 Chapter 2 (SC/ST Quotas) Annexure-I (Negative List of Ineligible Works).

[SCIENTIFIC & MATHEMATICAL REFERENCES]

* Liu, Ting, Zhou: Isolation Forest (IEEE ICDM) Multivariate anomaly detection.
* Newcomb (1881) Benford (1938): Forensic leading-digit logarithmic analysis P(d) log10(1 + 1/d).
* Cox, D.R. (1972): Regression Models and Life-Tables (Journal of the Royal) ' (Statistical Society) Survival hazard modeling.